Development Scope and Experimental Timeline of the

MAIA Framework

Authors: Iraê César Brandão (Developer) and Soumita Sarkar (Researcher/Creator)

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The MAIA (Model of AI Literacy) Framework is proposed as an experimental and analytical system aimed at the meta-evaluation and architectural validation of generative artificial intelligence models. It emerges in response to the growing reliance on large-scale language models within educational and research contexts, where the apparent diversity of systems may obscure underlying epistemological convergence.

Rather than evaluating isolated outputs in terms of correctness or performance, the MAIA Framework shifts the analytical focus toward the relational dynamics between multiple generative systems. In doing so, it enables the identification of patterns of convergence, redundancy, and divergence across outputs, thereby providing a basis for examining the structural behavior of distinct model architectures through their linguistic productions.

Central to this framework is the concept of **monocentric dependency**, here theorized as **"CHAT-GPTlogy"**, which refers to the tendency of different generative systems to reproduce semantically aligned responses, reinforcing a dominant epistemic structure while simulating diversity. This phenomenon raises critical concerns regarding the formation of knowledge, particularly in pedagogical environments where such systems are increasingly integrated.

To operationalize this analysis, the system introduces two core metrics: the **Monocentric Dependency Index (IDM)**, which quantifies the degree of convergence among model outputs, and the **Coefficient of Plurality (CPA)**, which measures the extent of divergence and potential epistemic variability. These indicators are not intended as definitive measures of truth or quality, but as heuristic tools for exploring relational patterns within AI-generated content.

The MAIA Framework is particularly situated within the domain of **ELT (English Language Teaching)**, where the integration of generative AI tools necessitates a critical approach to technological mediation in learning processes. By foregrounding plurality and encouraging comparative engagement with multiple systems, the framework seeks to foster a form of AI literacy grounded in epistemological awareness and methodological reflexivity.

Academic Note: *This tool is experimental and intended for research and prototyping in AI evaluation and literacy studies.*

Table 1. Development Phases and Technical Milestones

The development of the **MAIA Framework** (Model of AI Literacy) was conducted as an experimental scientific prototyping process between **January 2026 and March 05, 2026**. The project was designed as a browser-based analytical environment intended to support experimental validation, methodological exploration, and the meta-analysis of generative AI outputs in educational contexts.

Phase	Period	Activities Performed	Technical Justification
Phase 1 – Conceptual Planning	**Jan 02 – Jan 08, 2026**	**Definition of theoretical foundations, project naming, conceptual identity, logo design, and analytical objectives.**	**To establish the epistemological and methodological identity of the framework.**
Phase 2 – Development Environment Selection	**Jan 09 – Jan 12, 2026**	**Selection of Visual Studio Code as the main IDE.**	**Chosen for modular coding, extension support, debugging tools, and integrated version control.**
Phase 3 – Technology Stack Definition	**Jan 13 – Jan 18, 2026**	**Definition of HTML, CSS, and JavaScript as the core languages.**	**Chosen for browser compatibility, lightweight deployment, and full client-side execution.**
Phase 4 – Interface Architecture	**Jan 19 – Jan 28, 2026**	**Development of menus, input fields, tables, scatter plot areas, and responsive layout.**	**To ensure usability, intuitive navigation, and analytical interaction.**
Phase 5 – Mathematical Modeling	**Jan 29 – Feb 08, 2026**	**Implementation of similarity calculations, metric scoring (IDM/CPA), classification rules, and graphical plotting.**	**JavaScript arithmetic allowed dynamic real-time calculations without server dependency.**

Phase	Period	Activities Performed	Technical Justification
Phase 6 – Storage and Data Integrity	**Feb 09 – Feb 16, 2026**	**Implementation of local browser storage, cache persistence, XML/XLS export, XML import, and backup routines.**	**To preserve experimental sessions and support reproducibility.**
Phase 7 – Browser Validation	**Feb 17 – Feb 25, 2026**	**Cross-browser testing in Google Chrome, Microsoft Edge, Mozilla Firefox, and Opera.**	**To verify compatibility, rendering consistency, execution speed, and UI stability.**
Phase 8 – Experimental Validation	**Feb 26 – Mar 05, 2026**	**Prototype experimentation, performance testing, responsiveness validation, dataset generation, and scientific documentation.**	**To verify reliability, usability, analytical consistency, and academic reproducibility.**

Table 2. Technical Architecture and Design Decisions

The MAIA Framework was designed as a fully client-side web application, avoiding immediate dependency on external APIs or server-side infrastructures. This decision ensured unrestricted experimental execution, allowing researchers to test the prototype without requiring authentication tokens, API billing, or third-party user credentials.

The platform architecture included:

Component	Technology	Purpose
Interface Structure	**HTML**	**Layout, navigation menus, input forms, tables**
Visual Design	**CSS**	**Colors, typography, responsiveness, visual identity**
Logic & Arithmetic	**JavaScript**	**Similarity calculation, metric scoring, local storage, data export/import**
Repository & Deployment	**GitHub**	**Version control, code backup, public hosting**

Table 3. JavaScript Analytical Arithmetic

Calculation	Function
Word Overlap Similarity	**Measures lexical convergence between model outputs**
IDM Calculation	**Quantifies monocentric dependency**
CPA Calculation	**Measures plurality and divergence**
Weighted Classification	**Categorizes outputs as Monocentric, Hybrid, or Plural**
Scatter Plot Coordinates	**Dynamically plots analytical positioning**

This approach allowed real-time calculations and immediate graphical feedback during experimentation.

Table 4. Future Technological Prospects of the MAIA Framework (Automation Perspective)

The prototype may be automated in future versions through API integration with external AI systems. However, this was intentionally postponed during the initial development phase because such implementation would require user authentication tokens, API credentials, usage billing, and external processing dependencies, which could compromise free experimental accessibility and local reproducibility.

Future Expansion	Proposed Implementation	Scientific and Technical Purpose
Structured Database Integration	**Implementation of relational or non-relational databases such as MySQL, PostgreSQL, or MongoDB.**	**To improve long-term data persistence, scalability, analytical traceability, and multi-user research environments.**

Future Expansion	Proposed Implementation	Scientific and Technical Purpose
Cloud Infrastructure Deployment	**Migration to cloud environments such as Amazon Web Services, Google Cloud, or Microsoft Azure.**	**To enable distributed access, secure backups, computational scalability, and collaborative research deployment.**
Advanced Front-End Migration	**Adoption of frameworks such as React, Vue.js, or Angular.**	**To improve modularity, interface performance, dynamic rendering, and maintainability.**
Server-Side Architecture	**Backend implementation using Node.js, Python, or PHP.**	**To allow authentication, centralized validation, API communication, and advanced analytics.**
Token-Based AI Integration	**Integration with language model APIs through tokenized processing using platforms such as OpenAI, Anthropic, or Google DeepMind.**	**To automate response collection, semantic comparison, large-scale experimentation, and real-time model interaction.**
Semantic Analysis Enhancement	**Use of embeddings, vector databases, and transformer-based semantic models.**	**To move beyond lexical overlap and enable deeper conceptual similarity analysis.**
Automated Research Reports	**Generation of dynamic scientific reports in PDF, XML, CSV, and dashboard formats.**	**To increase reproducibility, documentation quality, and institutional research integration.**

Table 5. System Limitations

The current prototype presents the following methodological and technical limitations:

Limitation	Description
1- Simplified Similarity Method	**Uses lexical word overlap instead of deep semantic analysis**

Limitation	Description
2- Analytical Precision	Results may not fully represent conceptual similarity
3- Local Storage Dependency	Data may be lost if browser cache is cleared
4- No Server Validation	No backend persistence or external verification
5- Basic Visualization	Graphs are exploratory rather than statistically advanced
6- Manual Data Entry	User editing may introduce inconsistencies
7- Input Bias	The system does not automatically detect duplicated or biased entries

Future Implementation Considerations

Future versions of the MAIA Framework may incorporate token-based artificial intelligence integration, enabling automated interaction with external language models through API requests. This functionality was intentionally not implemented in the current prototype because tokenized systems require user credentials, API authentication, consumption quotas, and operational costs, which could restrict free experimental accessibility during the initial scientific validation phase.

Therefore, the current browser-based architecture prioritizes local execution, methodological transparency, low operational cost, and unrestricted academic experimentation, while maintaining scalability for future technological expansion.

Platform Architecture, Availability, and User Interaction of the MAIA Framework

1- Development Environment

The development of the MAIA Framework was carried out using the IDE Visual Studio Code, chosen for its modular environment, integrated debugging tools, code organization features, and compatibility with front-end web technologies. This environment enabled the structuring, testing, and iterative refinement of the prototype during all development stages.

2- Programming and Web Technologies

The system architecture was developed using three core web technologies:

- **HTML:** responsible for the structural composition of the web application, including menus, input forms, analytical tables, conceptual sections, and graphical containers.
- **CSS :** responsible for the visual identity of the platform, including typography, colors, spacing, responsive adaptation, and user navigation experience.
- **JavaScript:** responsible for the computational logic of the system, including arithmetic calculations, similarity metrics, dataset manipulation, local storage, export routines, and graphical rendering.

This architecture was intentionally designed as a **browser-based environment**, eliminating the need for server-side dependencies during the experimental phase.

3- Online Deployment and Scientific Availability

After the prototyping and validation phases, the source code was made publicly available through GitHub, a collaborative software development platform widely used by researchers and developers.

This decision was adopted for three main reasons:

- Scientific reproducibility : allowing other researchers to inspect and replicate the experimental environment;
- Version control: preserving development history and methodological updates;
- Collaborative scalability: enabling future improvements and external contributions.
- The experimental application is publicly accessible through:

MAIA Framework Prototype:

https://irae-cesar-brandao.github.io/maia_framework_prototype/

This deployment uses static web hosting through GitHub Pages, allowing direct browser execution without installation requirements.

4- Operational Method of Use

The MAIA Framework operates entirely within the user's browser. During use, the researcher manually inserts:

- The research prompt;
- Output generated by Model A;
- Output generated by Model B;
- Optional manual adjustments of IDM and CPA values.

Once inserted, the system performs local processing, calculates similarity scores, classifies the analytical profile, and plots the results on the scientific scatter plot.

During execution, the browser automatically creates temporary local cache and virtual memory structures, allowing continuity of experimentation and incremental data accumulation.

5- Navigation Structure and Menus

The user interface was organized into analytical modules:

- ***Home:*** *Presents the theoretical and epistemological foundations of the MAIA Framework, including the conceptual basis of AI literacy and monocentric dependency.*
- ***Framework:*** *Core experimental environment where prompts, model outputs, and analytical metrics are inserted and processed.*
- ***Dataset:*** *Displays all experimental records in tabular format, enabling comparative observation and editing of stored entries.*
- ***Graphic Results:*** *Generates the scientific scatter plot representing the relationship between **IDM** (convergence) and **CPA** (plurality/divergence).*
- ***Guide:*** *Provides methodological explanations, interpretation criteria, classification rules, and researcher-oriented analytical procedures.*
- ***Suggestions:*** *Offers complementary digital resources and educational tools related to academic English and AI literacy.*
- ***Disclaimer:*** *Presents methodological limitations, ethical considerations, and the academic scope of the prototype.*

6- Functional Buttons and Data Operations

The system also includes operational controls:

Function	Purpose
Save	**Stores a new experimental entry**
Delete Select	**Removes a selected dataset record**
Clear All	**Deletes all locally stored experimental data**
Export XLS	**Generates spreadsheet backup files**
Export XML	**Generates structured scientific data files**
Import XML	**Restores previously exported datasets**

7- Future Scalability

Although the current version operates locally, future versions may integrate databases, cloud services, and tokenized AI APIs. This was intentionally postponed in the current phase to avoid dependency on external credentials, token consumption, API billing, and authentication barriers, thereby preserving open academic experimentation.

Thus, the MAIA Framework currently positions itself as an open, browser-based scientific prototype, designed for iterative experimentation, conceptual development, and methodological research in AI literacy.

Development Process Conclusion MAIA Framework

Throughout the development period between **January and March 2026**, the MAIA Framework evolved from an initial conceptual proposition into a functional scientific prototype through a continuous process of theoretical reflection, technical experimentation, and collaborative decision-making. During this period, both authors maintained active and consistent participation in all development stages, including literature review, conceptual modeling, interface planning, methodological discussions, system validation, and experimental implementation.

The construction of the framework emerged from the shared identification of a research gap concerning **AI literacy**, epistemological plurality, and the comparative analysis of generative systems in educational contexts. From the initial conceptualization of the idea to its technical execution, the development process involved constant dialogue, critical evaluation of alternatives, and iterative refinement of both theoretical and computational components.

Technical decisions - such as the choice of the development environment, programming languages, interface architecture, analytical metrics, storage mechanisms, and deployment platform- were collectively discussed and validated according to criteria of scientific reproducibility, accessibility, and methodological transparency.

As a result, the MAIA Framework represents not only a technological artifact, but also the outcome of an interdisciplinary and collaborative research process, in which all authors contributed actively, consistently, and intellectually to the conceptual, methodological, and experimental maturation of the project. This collaborative engagement was fundamental to transforming an initial research idea into an operational prototype with academic applicability and future scalability.

Conflict of Interest

The authors affirm that there were no conflicts of interest related to the conception, development, experimentation, analysis, or publication of the MAIA Framework. All research activities were conducted independently, collaboratively, and exclusively for academic and scientific purposes, without financial, institutional, or commercial influence.

Conflict of Interest

This development scope consists of **nine (9) pages**, which formally conclude this stage of conceptual design, technical development, experimental validation, and methodological documentation of the MAIA Framework. The completion of this phase establishes the foundational structure for future scientific applications, technological expansion, and continued research development.